

Futuremark goes DX11

Futuremark Corp. is prepping a new DX11 benchmark, given the buzz around DX11. We look forward to benchmarking using that in future tests

New 125\$ SSD

Intel has launched their X25-V with a capacity of 40 GB priced at 125 dollars and aimed at entry-level budgets. Maybe we can all finally afford one

How stuff works

the water. This is something that isn't possible unless OIT is used.

This feature uses the Compute Shader to sort/arrange transparent textures in a single pass so that they can be rendered in the correct order to be displayed on-screen. This was possible using Pixel Shaders earlier, but resulted in a tremendous performance hit. With OIT onboard, you will now see a lot of realism in certain scenes and while many gamers will not have missed its benefits, once you have seen what a game can look like, there's no looking back. Texture sizes have also been bumped up, this stands at 16000 x 16000 pixels i.e. 256 megapixels — that's a lot!

Extreme gamers will also cheer, for Supersample AA (SSAA) makes a comeback. One of the reasons this mode of AA was left out in earlier cards was the huge performance hit. Therefore Multisample AA (MSAA) and Adaptive AA (AAA) were used but these had a few shortcomings. Thanks to the 2x processing capability, the HD 5xxx series feature SSAA. The HD 5xxx series also feature true Angle-Independent Anisotropic Filtering (AIAF) resulting in much improved visual quality with less artifacts and other visual anomalies.

AMD also built in support for up to 6 displays on each GPU. Although whether or not your card can do this depends on whether the card has six outputs. Support for 6 displays has also been built into the Catalyst driver. Even hooking up three displays is quite a treat as all the

display outputs can be configured into one single display surface, which to Windows, appears as a single monitor with a super high resolution. Gaming on such a set up is immense fun as you can well believe.

Performance-per-watt anyone?

This was a phrase that AMD used a lot for their CPUs. However, power consumption is becoming more of a factor for GPUs especially given they consume three times the power of today's CPUs. Although power consumption on load has gone up on the HD 5870, but it is a marginal increment over the HD 4870, and the performance increment is much higher aka more performance per watt. Interestingly, thanks to power saving features implemented, the power draw on idle is down to just 27 watts, this was in excess of 80 watts on the HD 4870. As a result of this, as well as the 40nm fabrication process, the HD 5870 is a cool running card, you might even say its tropicalised!

It's good to be a gamer right now

DX11 was designed as a lighter DX10 with a few visual enhancements and a lot of features aimed at making development on this platform easier. We've already tested cards based on nearly all AMD 5xxx series of GPUs and we're happy to report that AMD has a solid



Notice the different transparent layers and how one overlaps the other. The order of arranging the layers is important for realism, this is what OIT handles — which is the top layer, which is just beneath that and so on

DX11 line-up. The performance of the HD 5870 is sufficiently incremental to the HD 4870 for us to recommend it, even if DX11 were to be taken out of the equation and a similar scenario exists for the entire lineup. And the sales figures corroborate our assumption as AMD has enjoyed a good last six months. But *the times are a changin'* to borrow a cliché and NVIDIA is very close to a market release of their own DX11 card. They've announced the GTX 480 and GTX 470 cards and we've taken a look at this GPUs innards last month in a similar article. Needless to say — it's quite impressive, at least on paper for the time being. So the future of PC gaming is looking very bright, a suitable snub to all doomsday prophesiers and PC haters alike. There's a lot to be hopeful for, if you're a PC gamer — DX11 games are slowly trickling out of studios around the world, we've played some of them and are most impressed. And then NVIDIA is joining the fray shortly. Hopefully the next few months will see some much needed competition to get prices down to bare minimum levels. 



Eyefinity with 6 monitors — A new level of immersion